12.6 Conditional Probability

Illustration 12.8

Consider the following example to understand the concept of conditional probability.

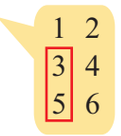
Suppose a fair die is rolled once, then the sample space is $S = \{1, 2, 3, 4, 5, 6\}$. Now we ask two questions

Q_1: What is the probability of getting an odd number which is greater than 2?

Q_2: If the die shows an odd number, then what is the probability that it is greater than 2?

Case 1

The event of getting an odd number which is greater than 2 is $\{3, 5\}$.



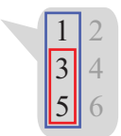
Let P_1 be the probability of getting an odd number which is greater than 2

$$P_1 = \frac{n(\{3,5\})}{n(\{1,\ 2,\ 3,\ 4,\ 5,\ 6\})} = \frac{2}{6} = \frac{1}{3}.$$

Case 2

'If the die shows an odd number' means we restrict our sample space S to a subset containing only odd number.

That is, $S_1 = \{1, 3, 5\}$. Then our interest is to find the probability of the event getting an odd number greater than 2. Let it be P_2



$$P_2 = \frac{n(\{3,5\})}{n(\{\ 1,\ 3,\ 5\})} = \frac{2}{3}$$

In the above two cases the favourable events are the same, but the number of exhaustive outcomes are different. In case 2, we observe that we have first imposed a condition on sample space, then asked to find the probability. This type of probability is called conditional probability.

This can be written by using sample space as

$$P_2 = \frac{\frac{n(\{3,5\})}{n(\{1,2,3,4,5,6\})}}{\frac{n(\{1,3,5\})}{n(\{1,2,3,4,5,6\})}} = \frac{\frac{2}{6}}{\frac{3}{6}} = \frac{2}{3}$$

Important note: Sample space is same for probability and conditional probability.

Definition 12.14

The **conditional probability** of an event B, assuming that the event A has already happened is denoted by $P(B/A)$ and is defined as

$$P(B/A) = \frac{P(A \cap B)}{P(A)}, \quad \text{provided } P(A) \neq 0$$

Similarly,

$$P(A/B) = \frac{P(A \cap B)}{P(B)}, \quad \text{provided } P(B) \neq 0$$

Example 12.16

If $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$

Find (i) $P(A/B)$ (ii) $P(\overline{A}/B)$ (iii) $P(A/\overline{B})$.

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